

Unit vector flows

OPG-136

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Conjecture For every graph G without a bridge, there is a flow $\phi : E(G) \rightarrow S^2 = \{x \in \mathbb{R}^3 : |x| = 1\}$.

Conjecture There exists a map $q : S^2 \rightarrow \{-4, -3, -2, -1, 1, 2, 3, 4\}$ so that antipodal points of S^2 receive opposite values, and so that any three points which are equidistant on a great circle have values which sum to zero.